USER GUIDE

ZEUS ver0.4.0

13/08/2026

[GitHub]

ZEUS's General GUI Layout

ZEUS GUI contains a progress bar at the top followed by a 3-panel setup as shown below. The left panel contains settings for the specific step, the top right panel will always be either a Nyquist or a Bode plot, and the bottom right panel will display the step-specific plot. Below the three panels, there is a status update to show if a step has been completed.

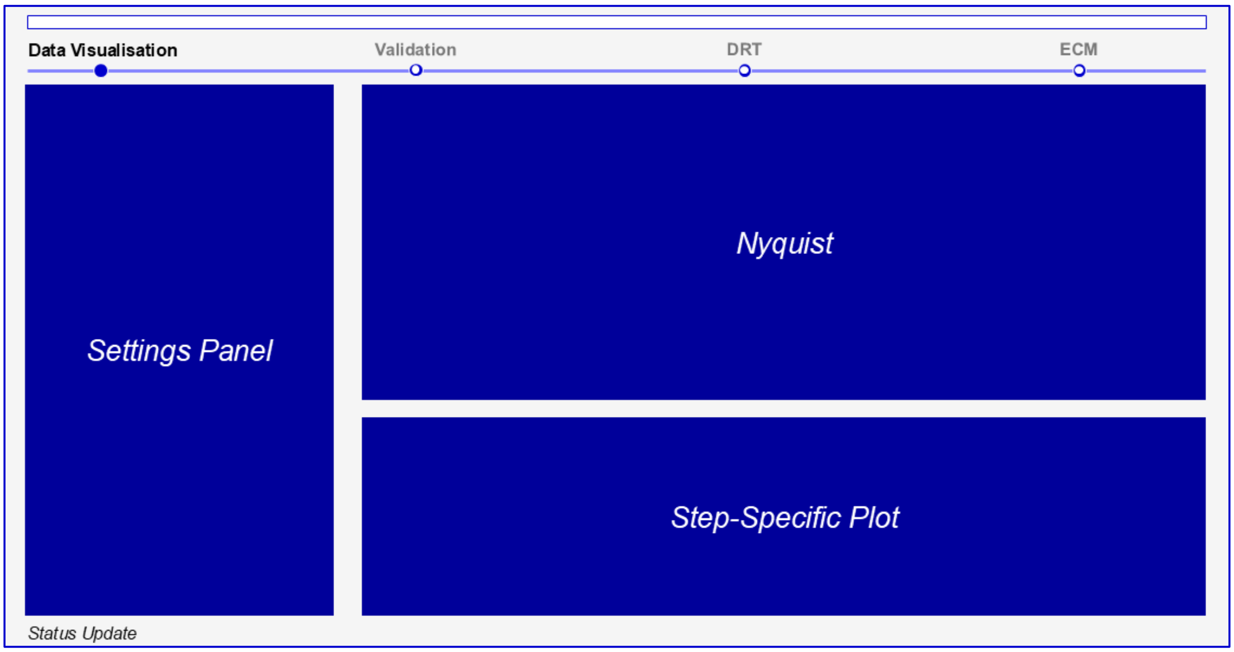

Figure 1: ZEUS General GUI Layout

ZEUS's Workflow

ZEUS follows a modular workflow where the user can decide which combination of the four steps below to perform in their analysis.

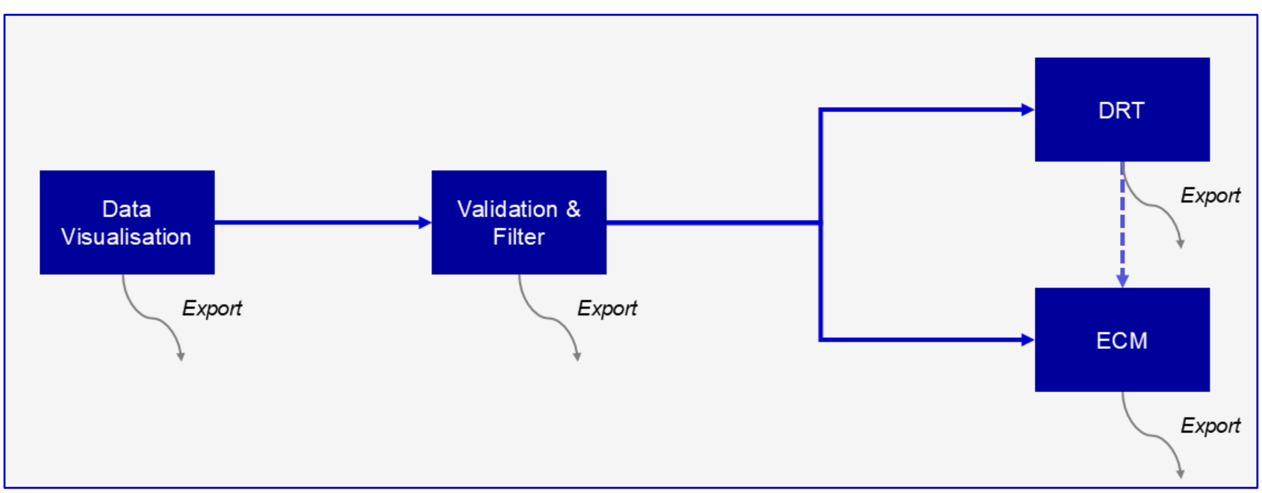

Figure 2: ZEUS's Workflow

Import/Load Data

1) Upon launch, ZEUS will land on the ***Data Visualisation*** step.
2) Load data:
   a) Click ***Clear All and Add Files...*** or ***Add Files...*** and pick one or more .mpt, .txt, or .csv files.
      i) For a generic .txt/.csv file, a popup dialog will ask you to confirm which column is frequency, Z', Z'', etc as shown in Figure 3 .
      ii) .mpt files containing multiple PEIS/GEIS sweeps (e.g. Modulo Bat sequences) are parsed automatically.
   b) To continue a previous analysis, use ***File → Open session...*** and pick a previously saved .eisz session file. This restores the loaded sweeps along with any validation/DRT/ECM results and filter state already computed.
3) Use the ***Active Dataset*** list on the bottom right section to select which spectra to work on as illustrated by the orange box in Figure 3.
4) Adjust marker shape, marker size and line width if needed in the ***Plot Options***.

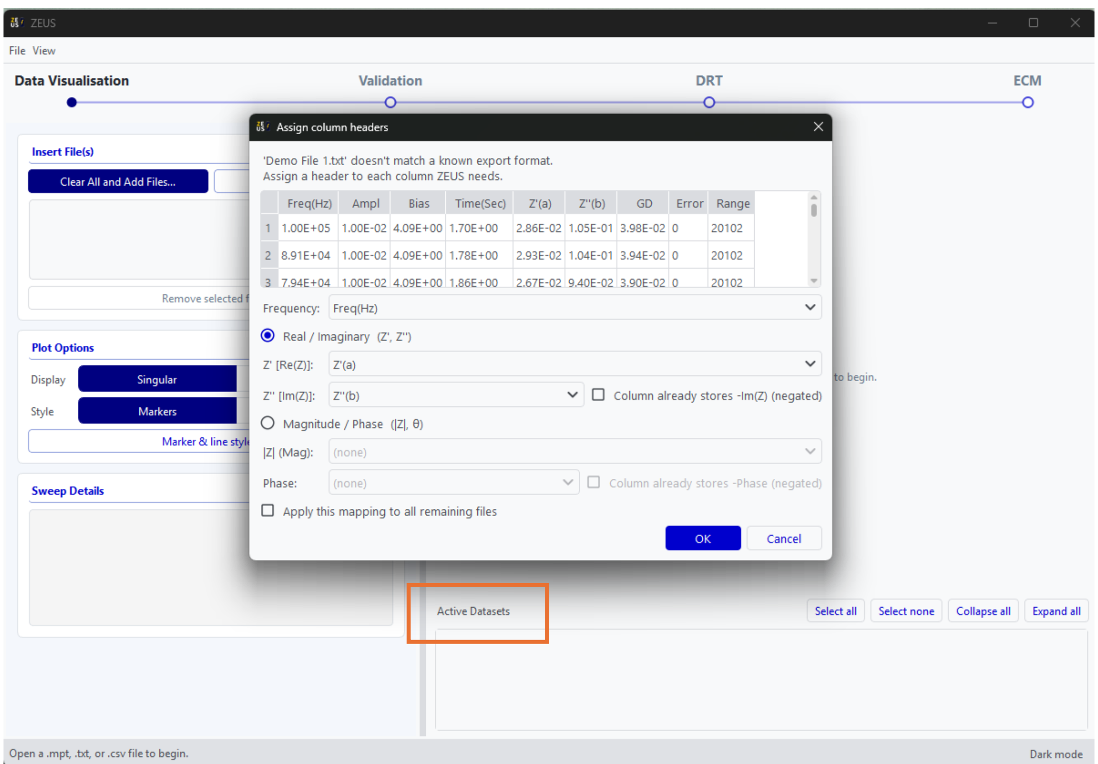

Figure 3: Data Visualisation Tab

Validation

1) Move to the ***Validation*** step by clicking on it in the progress bar.
2) Removing inductive tail is recommended as ZEUS works best without inductive processeses.
3) Choose between ***Kramers-Kronig*** or ***Z-HIT*** to see how consistent each spectrum is with a linear, causal, stable system.
4) Choose to define residual as either *ΔZ/|Z|* or *ΔZ'/Z'*.
5) Choose from either ***Basic*** or ***Advanced*** threshold mode:
   a) ***Basic***: One hard limit. Any point above this limit will be removed.
   b) ***Advanced***: One hard limit and one soft limit. Similarly, any point above the hard limit will be removed. On top of that, the *worst* point between the soft and hard limit will be discarded. Validation is rerun on the remaining point and the step is repeated until all points fall below the soft limit or until a maximum number of points have been removed to avoid ZEUS from crashing.
6) Review the residuals plot. Users can scroll through different sweeps using the arrow buttons. If there are points to be removed manually, user can do so by using the *eraser* tool inside the plot as shown by the orange box in Figure 4.
7) Once validation has been performed, the data is masked and will be carried forward automatically into DRT or ECM as the user chooses.

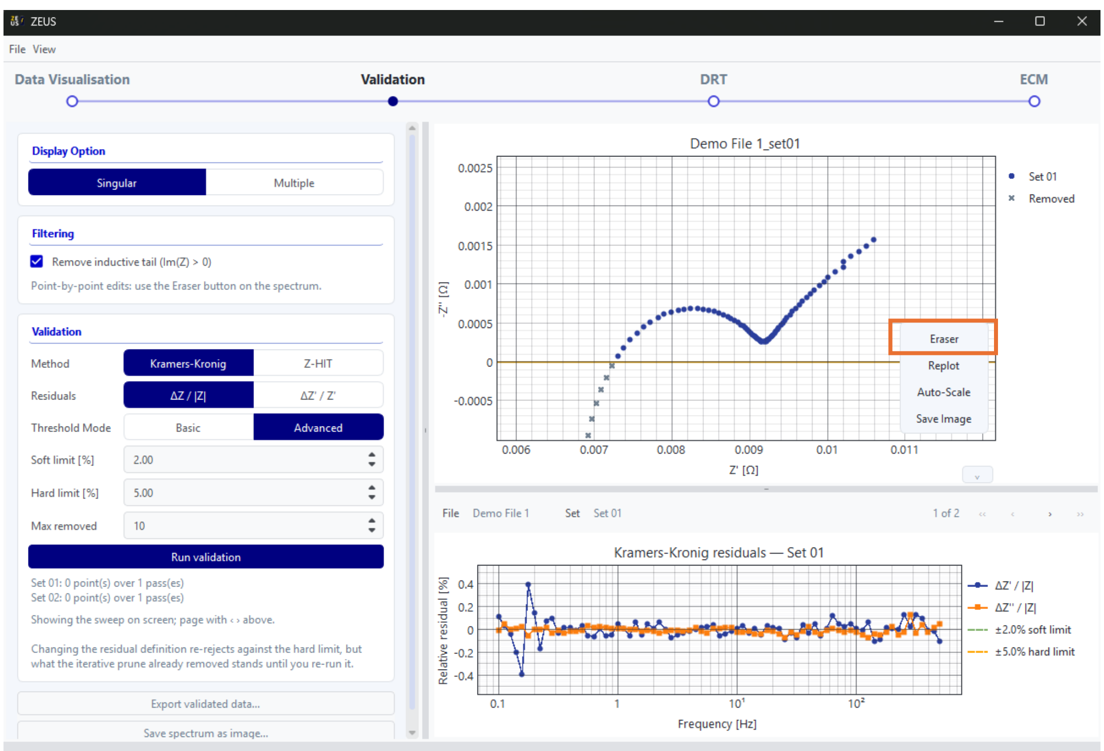

Figure 4: Validation Tab

Compute DRT

1) Move to the **DRT** step.
2) Choose a DRT method and basis function. Any irrelevant settings will be greyed out depending on which method is chosen. DRT is typically fitted to **Re + Im** but users may choose to only fit it to either one. Subtraction of diffusion tail is possible, however this option is still in BETA as shown by the orange box in Figure 5. Regularisation and RBF shape controls the appearance of the DRT plot. Very low regularisation parameter ,λ, will introduce a lot of ripples/noise while very high λ carries the risk of flattening the DRT peaks. On the other hand, RBF shape control sets how wide each basis function is: a narrower width resolves closely-spaced peaks more sharply but admits more noise, while a wider width smooths the distribution and can merge nearby peaks. Sampling is a Bayesian-specific setting which sets the number of HMC simulations.
3) Once DRT is performed, a DRT plot will appear in the bottom right corner. Users can then scroll through the different sweeps or click **Multiple** in **Display Option** to view all sweeps superimposed as shown in Figure 6.

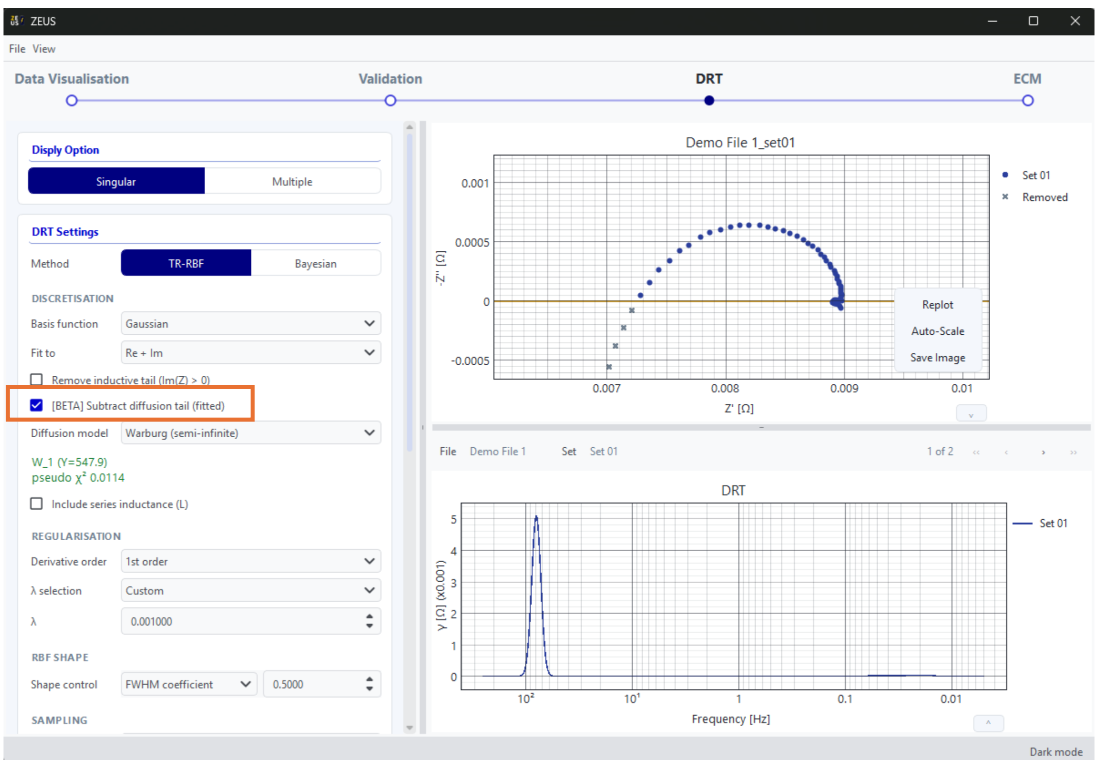

Figure 5: DRT Tab (Part 1)

4) Peak extraction collects and stores DRT information, which could be used for ECM. By default, Peaks is set to 0. This means that ZEUS will identify any local maximum as a peak, which could result to overfitting. In this current release, users will have to manually

change Peaks to the number of peaks they see from the DRT curve as shown by the orange box in Figure 6. As a result, ZEUS will extract the largest peaks corresponding to the number selected by the user.

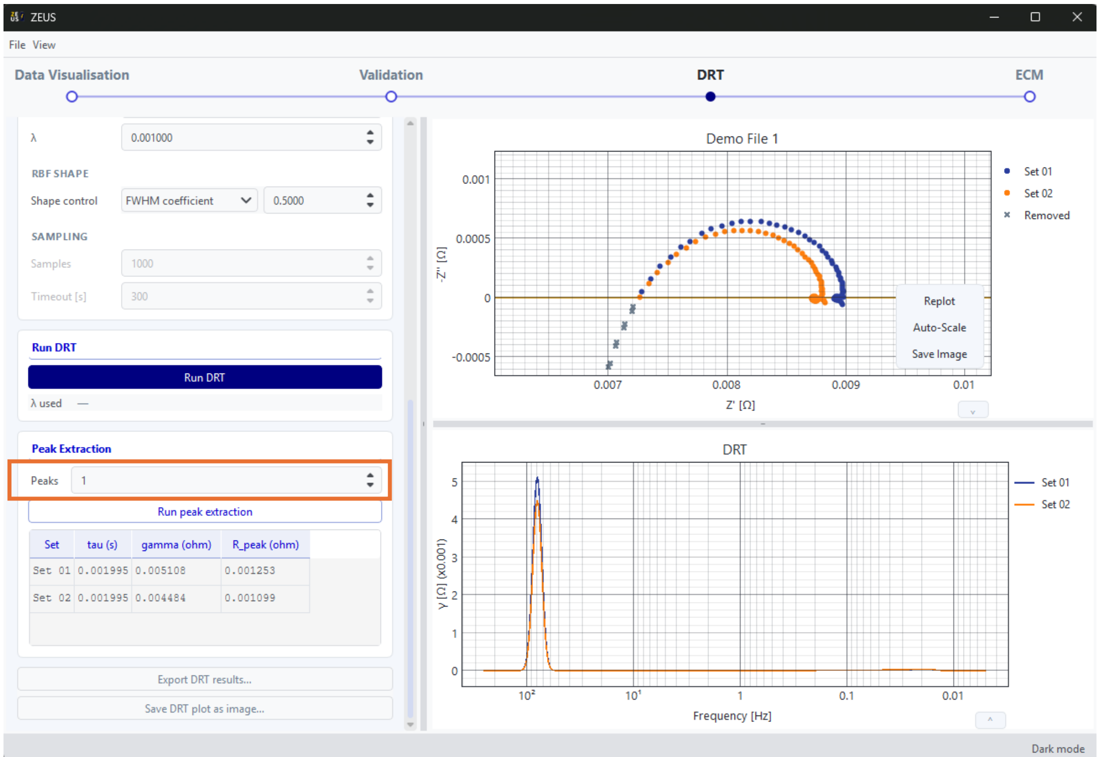

Figure 6: DRT Tab (Part 2)

ECM

1) Move to the **ECM** step.
2) Either:
   a) Click to seed a circuit from the DRT peaks found in the previous step illustrated by the orange box in Figure 7, or
   b) Enter a circuit description code (CDC) manually/build the circuit visually using the circuit diagram editor.
3) Choose a combination of **Method** and **Weight** which specifies the fitting method and the weightage of each point in the sweep respectively. By default, *least squares method* and *Boukamp weightage* are chosen.
4) Run the fit and inspect the fitted parameters, their uncertainties, and the residuals between the fit and the measured spectrum.

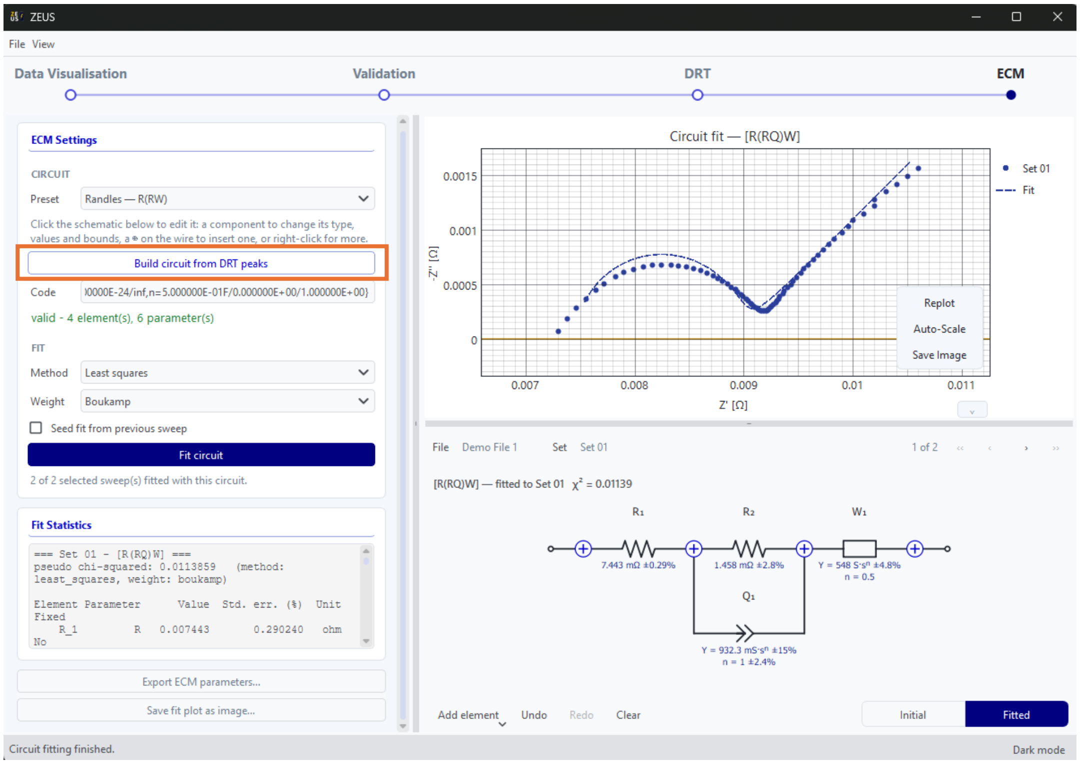

Figure 7: ECM Tab

Save and export

- ***File → Save session...*** saves everything you've loaded and computed (sweeps, validation/DRT/ECM results, filter state) to a single .eisz file.
- Each step also has its own export controls (e.g. validated data, DRT results, ECM parameters) for ZView-compatible output.
- Plots can be saved directly as image files from their right-click/toolbar menu.